

Mapping wireless networks worldwide

Wigle.net is a unique platform that consolidates location and other data on wireless networks around the globe. This data is crowdsourced from volunteers who use an app on their phones to log all the access points (APs) they encounter along with their GPS coordinates. This collected information is then fed into the Wigle database, creating a comprehensive map of wireless networks that is accessible through an easy-to-use website and app.

The process of data collection on Wigle.net relies on the community. Volunteers download the Wigle app on their smartphones, which automatically logs any APs they encounter, recording the SSID and GPS coordinates. This crowd-collected data is then aggregated into the Wigle database, allowing users to explore wireless network data from all over the world.

A common use case for Wigle.net is locating a target by their tethered AP SSID. For instance, many people use their phones as tethering APs (hotspots) when they need a Wi-Fi connection in a remote location or when available Wi-Fi is slow or expensive. If this has been done, chances are Wigle has recorded it.

At EFY, this API has been utilised to develop a DIY device focused on SSID-based location tracking technology. While the device used Wi-Fi, a similar device can be implemented using Bluetooth, although Bluetooth has a shorter range compared to Wi-Fi. This technology and API open possibilities for various other innovative devices.

What next?

The development of chips utilising SSID-based Wi-Fi locationing technology has significantly advanced, with some companies successfully

creating such solutions. This technology's resilience to signal disruption and fading makes it an invaluable enhancement to traditional global navigation satellite systems (GNSS). Looking ahead, the future is likely to see more companies adopting and expanding upon this technology, further improving the accuracy and reliability of location-based services.

A significant step forward in this field is the introduction of a new Wi-Fi companion IC (integrated circuit) by Nordic Semiconductor, which enhances Wi-Fi network scanning capabilities. When paired with a cellular IoT system-in-package (SiP), this IC forms a comprehensive silicon-to-cloud locationing solution. This integration covers Wi-Fi, cellular IoT, and GNSS, demonstrating the versatility and adaptability of SSID-based Wi-Fi locationing for both indoor and outdoor applications. This combination is especially useful in areas where GNSS signals are weak or absent.

The nRF7000 IC is designed for SSID-based Wi-Fi locationing, excelling in both active and passive scanning of Wi-Fi access points to enable precise positioning and location awareness. While it does not support direct Wi-Fi communication for internet connectivity, it integrates seamlessly with cellular SiPs. This allows developers to utilise GNSS, cell-based, and SSID-based Wi-Fi locationing for robust and accurate location results, while using the SiP for cloud connectivity via cellular networks.

The active and passive scanning capabilities of the IC allow detailed scanning of Wi-Fi networks, significantly enhancing location accuracy in areas with a high density of Wi-Fi access points. This combination of Wi-Fi and cellular technologies opens new possibilities for IoT applications that require reliable and accurate location information,

alongside internet and cloud connectivity through the SiP. Additionally, the IC supports hosted devices, broadening its applicability.

At present, only a few chips feature advanced SSID-based Wi-Fi locationing, but this is expected to change as more companies enter the market, improving tracking and navigation systems. The ongoing development and integration of SSID-based Wi-Fi locationing with other technologies, such as GNSS and cellular IoT, promise significant advancements in precise location tracking.

SSID-based Wi-Fi locationing marks a major advancement in location technologies, particularly in challenging environments where traditional methods like GPS struggle. By utilising the ubiquitous presence of Wi-Fi networks, this technology provides an effective and energy-efficient solution for accurate location tracking, especially in indoor and urban settings. Its integration into everyday devices enhances real-time navigation and opens new possibilities for indoor navigation, asset tracking, and personalised services. As this technology continues to develop and integrate with other data inputs and systems, it promises to deliver even greater accuracy and utility, making it an increasingly valuable tool in the arsenal of location-based services. This marks a shift from conventional location systems, introducing more advanced and reliable tracking solutions for modern industries. **EFY**

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